



Pre-Junior Cycle Final Examination, 2024

Mathematics

Ordinary Level

Time: 2 hours

270 marks

CANDIDATE DETAILS

**DAY and DATE
of BIRTH**



____ / ____

For example, 3rd February
is entered as 03/02

NAME

[illegible]

SCHOOL

[illegible]

TEACHER



School Stamp

For Examiner only

Question

Mark

1.

2.

3.

4.

5.

6.

7.

8.

9.

10

11.

12

Total

%



Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. You may ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

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(Suggested maximum time: 5 minutes)

- (i) $376 + 421$

[illegible]

- [illegible]

- | | | | |
|-------|-------------|---------|---------------|
| $-1,$ | $\sqrt{3},$ | $1.57,$ | $\frac{1}{2}$ |
|-------|-------------|---------|---------------|

Answer =		,		,		,	
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-

- Answer =

Question 2 (Suggested maximum time: 10 minutes)

Mark is driving from Dublin to Liverpool. He takes the ferry from Dublin to Holyhead and then continues his drive to Liverpool. The table below shows the times the ferry departs Dublin and arrives in Holyhead on a particular day.



Departing	Arriving
Dublin 02:00	Holyhead 05:25
Dublin 08:05	Holyhead 11:30
Dublin 08:15	Holyhead 11:50
Dublin 14:30	Holyhead 18:00
Dublin 14:45	Holyhead 18:20

- (a)** Mark needs to arrive in Holyhead by 5:00 pm.
What is the departure time of the latest ferry he can take from Dublin?

Answer =

- (b)** Work out the time this ferry will take in hours and minutes.

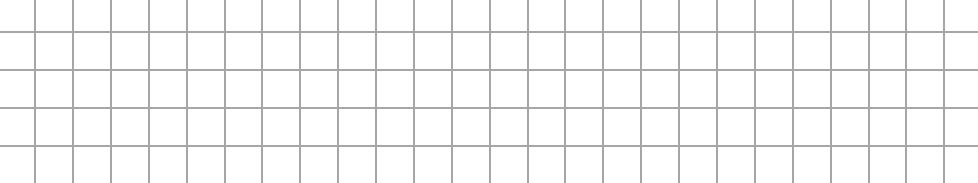


- (c)** Mark needs to arrive at Dublin port 30 minutes before the ferry leaves.
Work out the latest time that Mark needs to be at the port to ensure he gets on the ferry.

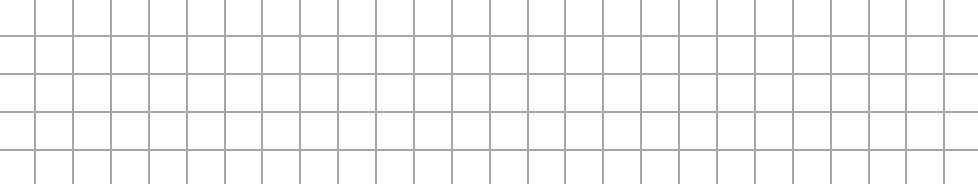
Answer =

[illegible]

- (d)** The distance from Holyhead to Liverpool is 150 km.
Mark drives at an average speed of 75 km per hour.
How long does the car journey from Holyhead to Liverpool take?



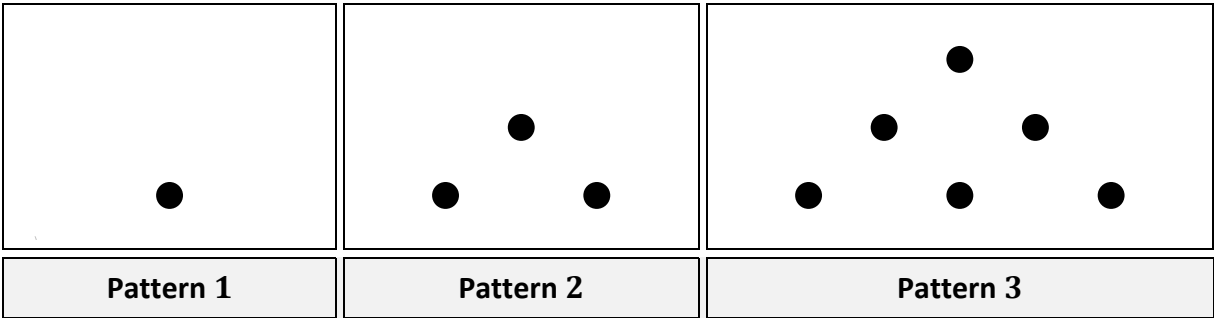
- (e)** Mark stays at a hotel in Liverpool for three nights. It costs him £250.
The currency conversion rate at the time is €1 = £0.88.
Convert £250 to euro (€). Give your answer correct to the nearest cent.



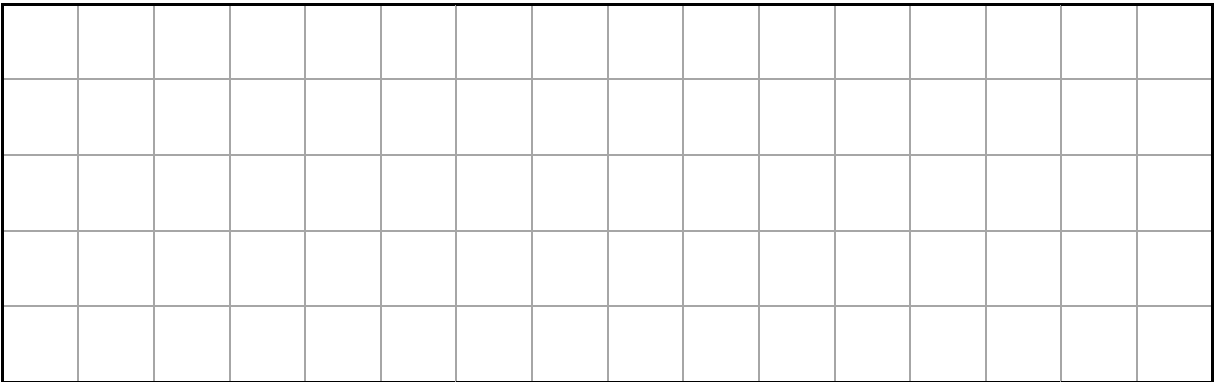
Question 3

(Suggested maximum time: 10 minutes)

Counters are placed in positions to make a sequence of patterns.
The first three patterns in a sequence are shown below.

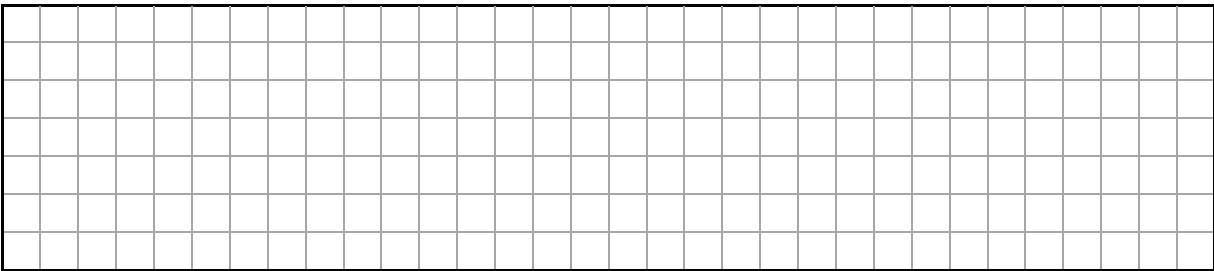


(a) Draw **Pattern 4** in the sequence.



(b) Fill in the table below to show the number of **counters** in each of the first five patterns.
One is already done for you.

Pattern	Number of Counters
1	
2	3
3	
4	
5	



- Tick (✓) **one** box only.

non-linear

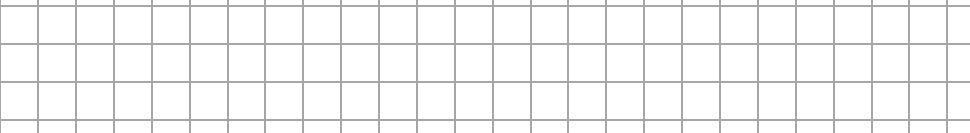
1

Reason:

- (d)** The number of counters in Pattern n is:

$$\frac{1}{2}(n^2 + n)$$

Use this to find the number of counters in **Pattern 10** (when $n = 10$).



Question 4 (Suggested maximum time: 10 minutes)

Question 4 (Suggested maximum time: 10 minutes)

- (a)** Simplify $b - 2a + 3b - 3a$.

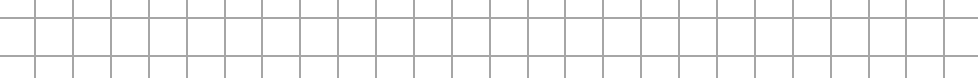
- (b)** Factorise $a^2 - 81$.

[illegible]

- (c) Fill in the table below, writing an algebraic term, in terms of N , to match the description. One has already been filled in for you.

Description	Algebraic term
Alex's age.	N
Alex's age in 6 years' time.	
Alex's age 3 years ago.	
Triple Alex's age.	
Half of Alex's age.	

- (d)** Alex is 14 years old. Lily is half Alex's age.
Work out how old Lily will be in 8 years' time.

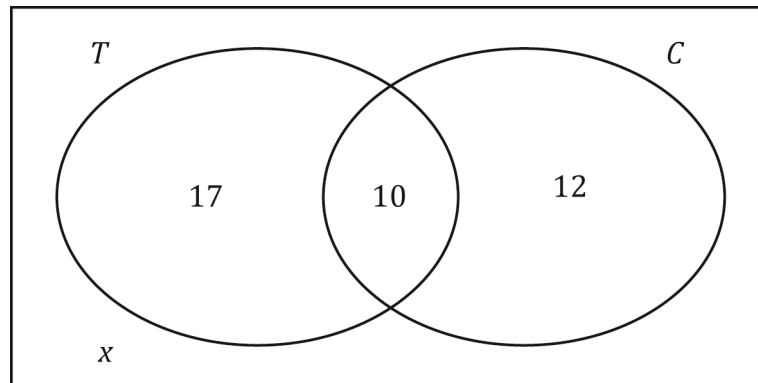


Question 5 (Suggested maximum time: 5 minutes)

Question 5 (Suggested maximum time: 5 minutes)

In a survey, a group of students were asked if they drink tea (T) or coffee (C).

The results are shown in the Venn diagram below, where $x \in \mathbb{N}$.



- (a)** How many students in the group drink tea?

[illegible]

- (b)** Explain what the following statement means, in terms of the students in the group:

$\#C = 22$.

[illegible]

- (c) 46 students took part in the survey.

What is the correct value of x , the number of students who drink neither tea nor coffee?

Show your working out.

Tick (✓) **one** box only.

$$x = 4$$

9

$$x = 7$$

9

 $x = 11$

9

Working out:

[illegible]

(Suggested maximum time: 10 minutes)

(a) The following terms can be used to describe the probability that an event happens.

Impossible	Unlikely	Even Chance	Likely	Certain
------------	----------	----------------	--------	---------

For each event **A**, **B** and **C** in the table below, use one of these terms to describe the probability that it happens.

Event	Description	Probability
A	Getting a number greater than 6 on a fair die.	
B	Getting a head when you toss a fair coin.	
C	Picking a day of the week that ends in a y .	

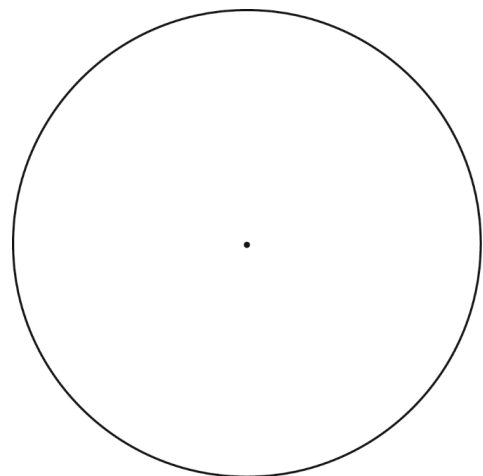
(b) It rains in Inverness in Scotland, on average, for 20 days during the month of April.

(i) While in Scotland on holiday during the month of April, Sophie travels to Inverness for a day.
Write down the **probability** that it rains while she is there.

Hint: There are 30 days in April.

[illegible]

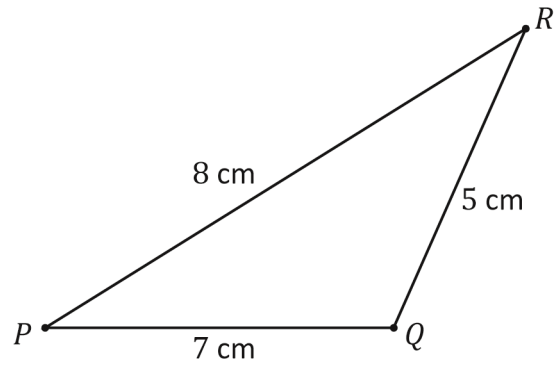
(ii) Complete the pie chart below to compare the numbers of wet days and dry days in Inverness during April. Clearly label each sector **and** indicate the size of each angle.

A full-page sheet of white graph paper with a light gray grid. The grid consists of small squares, approximately 1 cm by 1 cm each. There are 20 columns and 20 rows of squares. A thicker black border runs along the top and left edges of the page.

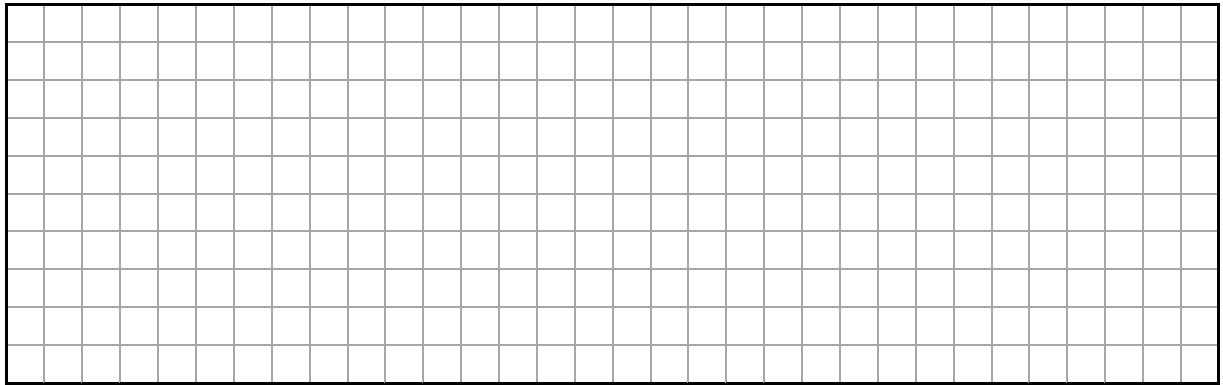
Question 7

(Suggested maximum time: 10 minutes)

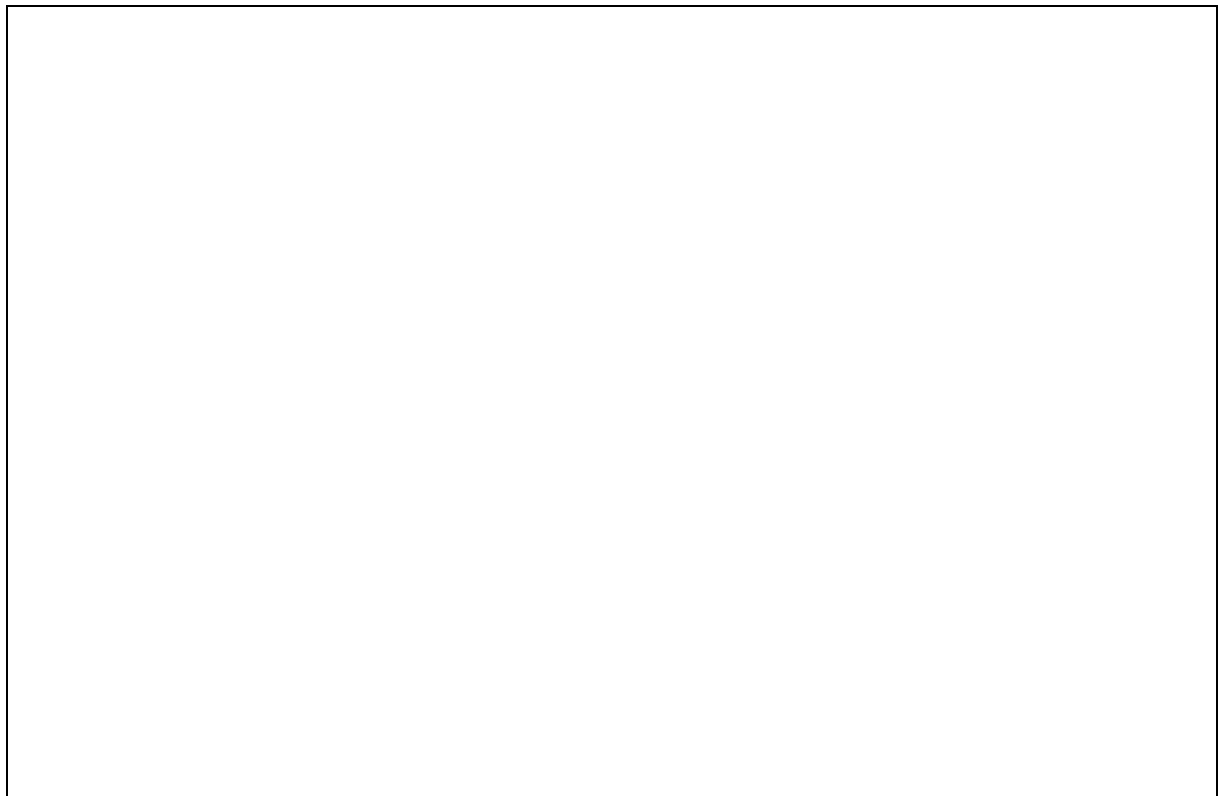
A triangle PQR has sides of length 7 cm, 8 cm and 5 cm as shown in the diagram (not to scale).



- (a) Work out the length of the **perimeter** of the triangle PQR .
Give your answer in cm.



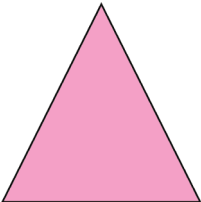
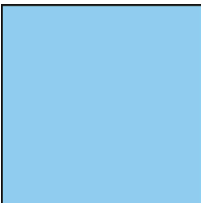
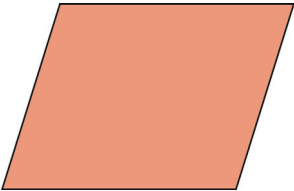
- (b) **Construct** the triangle PQR in the space below.
Show all of your construction lines clearly.



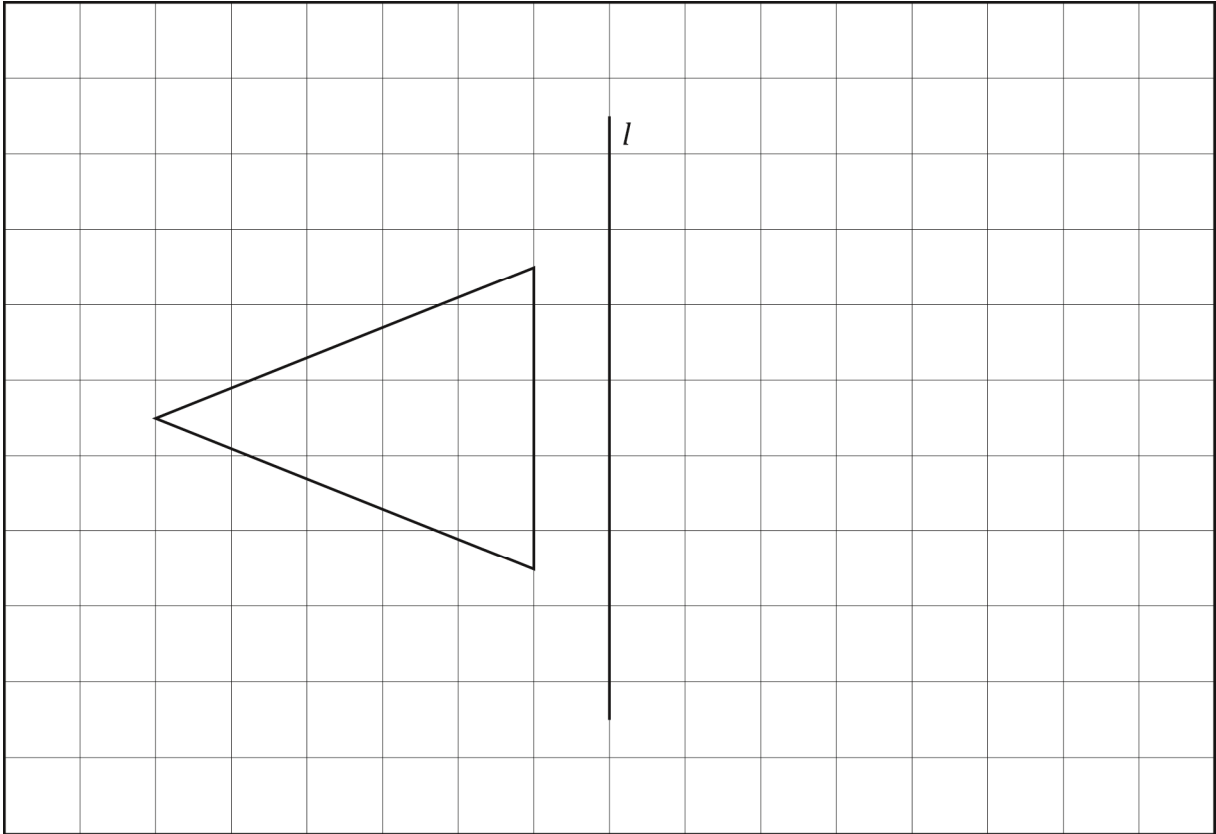
Question 8

(Suggested maximum time: 5 minutes)

- (a) The table below shows a number of shapes.
Fill in the table to show the number of axes of symmetry in each shape.

Shape	Diagram	Number of axes of symmetry Tick (✓) one box only.
Isosceles Triangle		☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4
Square		☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4
Parallelogram		☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4

- (b) Draw the image of the following triangle under **axial symmetry** in the line l .

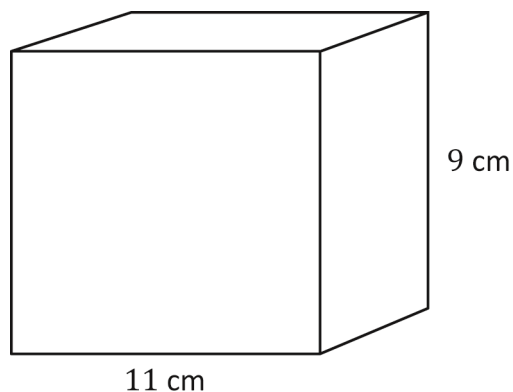
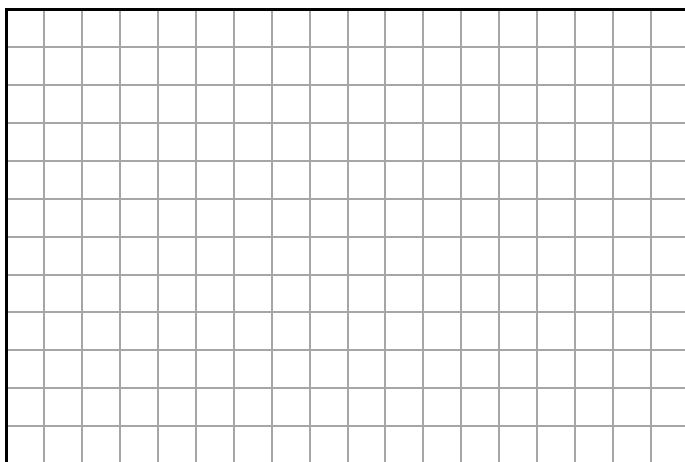


Question 9

(Suggested maximum time: 10 minutes)

A closed rectangular box has a height 9 cm, and a square base with sides of length 11 cm.

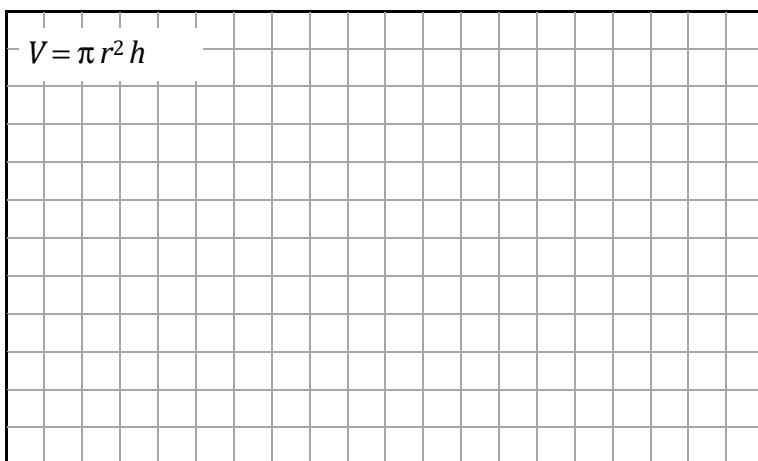
- (a) Work out the **volume** of the box in cm^3 .



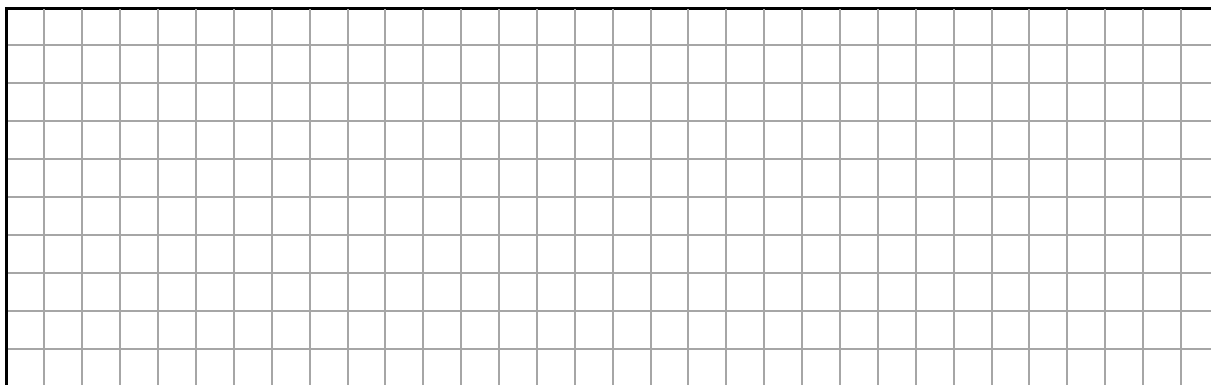
A household smart speaker is packaged in this box.

The smart speaker is in the shape of a cylinder with a height of 8 cm and a radius of 5 cm.

- (b) Work out the **volume** of the smart speaker.
Give your answer correct to the nearest cm^3 .



- (c) The remaining space in the rectangular box is filled with padding to protect the smart speaker during transport. Find the **volume** of padding required to fill this space.



(Suggested maximum time: 10 minutes)

A close-up photograph of a white golf ball resting on a green lawn. To the right of the ball is a circular hole in the grass, with a white flagstick partially visible. The background is a blurred view of a golf course under bright sunlight.

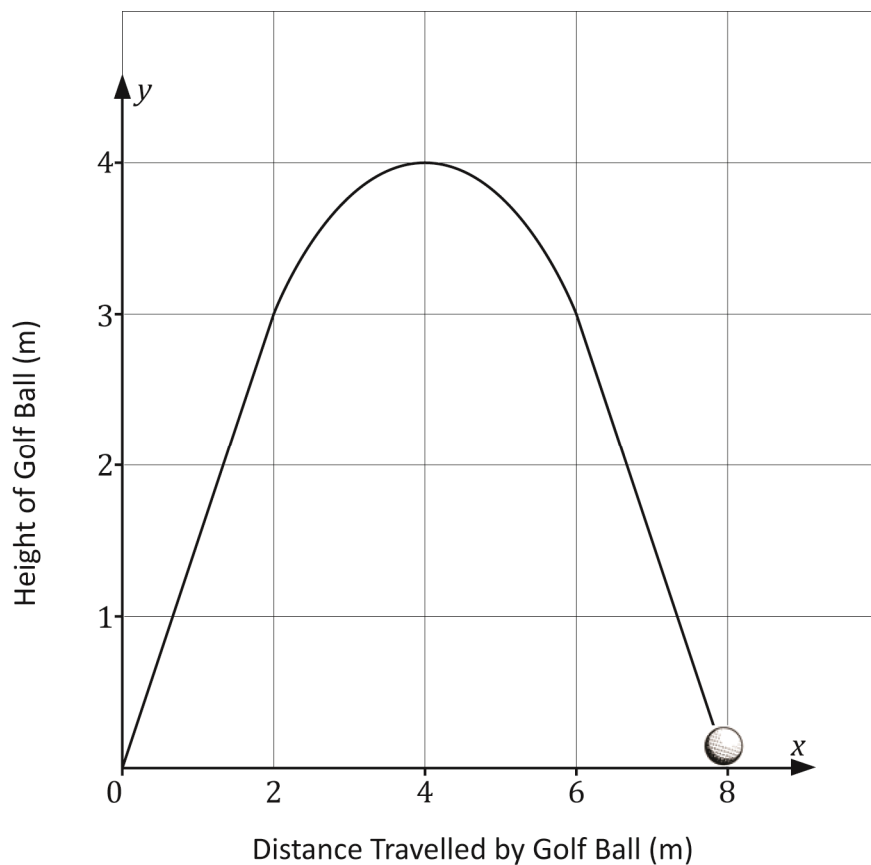
Number of golf shots taken	60 – 69	70 – 79	80 – 89	90 – 99
Number of golfers	2	6	11	8

- [illegible]

- | |
|--|
| |
|--|

-
- A blank coordinate plane with the x-axis labeled "Number of Golf Shots Taken" and the y-axis labeled "Number of Golfers". The x-axis has major tick marks at 60, 70, 80, 90, and 100. The y-axis has major tick marks from 1 to 12. A grid is drawn with vertical lines every 10 units on the x-axis and horizontal lines every 1 unit on the y-axis.

- (d) A flop shot is a speciality golf shot designed to hit the golf ball high and let it land softly. The diagram below shows the path of the ball during one of these shots.



- (i) Use the diagram to estimate the **height** of the ball when it had travelled a distance of 6 metres.

Height = m

- (ii) Use the diagram to estimate **distance** travelled by the ball when it had reached a height of 4 metres.

Distance = m

Question 11

(Suggested maximum time: 10 minutes)

- (a) For each statement below, tick (✓) the correct box which makes the statement true.
In each case, tick **one** box only.

- (i) The **sum of the three angles** in every triangle is

90°

☐

360°

☐

180°

☐

- (ii) An **acute** angle is

greater than 90°

☐

less than 90°

☐

greater than 180°

☐

- (iii) In an isosceles triangle, the number of sides that are equal in length is:

none

☐

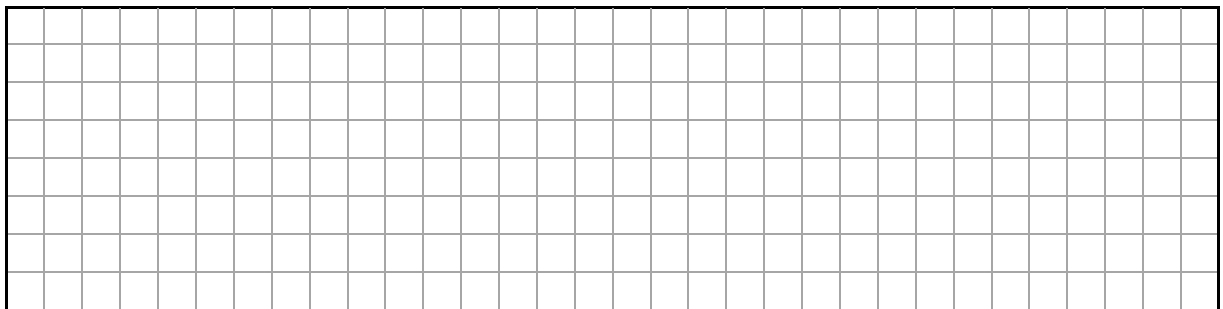
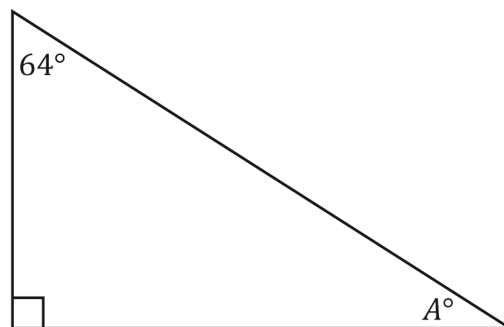
2

☐

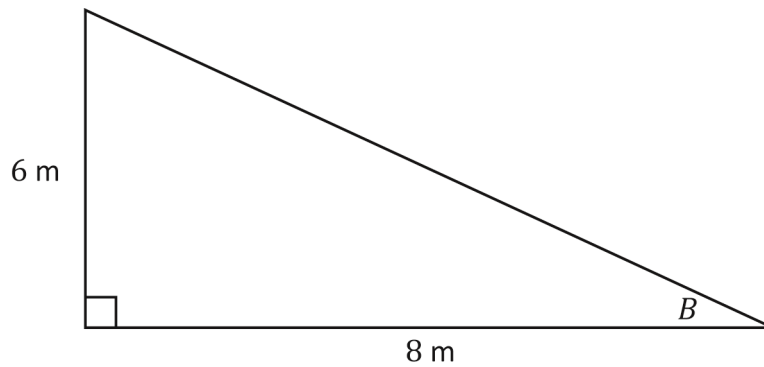
3

☐

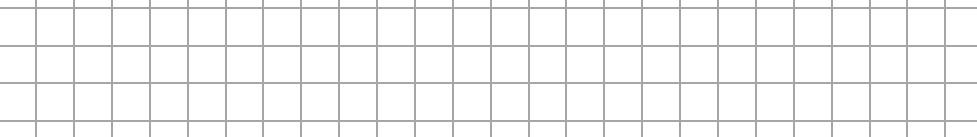
- (b) Work out the size of the angle marked A in the right-angled triangle shown (not to scale).



- (c)** The diagram shows another right-angled triangle (diagram not to scale).



- (i) Use the theorem of **Pythagoras** to find the length of the hypotenuse. Show all of your working out.



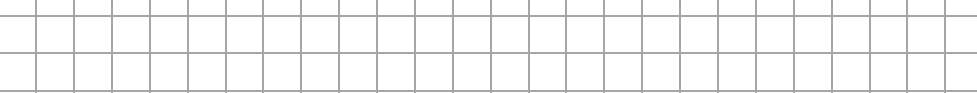
- (ii) Write down the length of the side **adjacent** to angle B .

Adjacent side = m

- (iii) Use the right-angled triangle in the diagram to write $\tan B$ as a fraction.

$$\tan B =$$

- (iv)** Use your answer from part **(c)(iii)** and your calculator to find the size of angle B . Give your answer correct to 2 decimal places.



Question 12

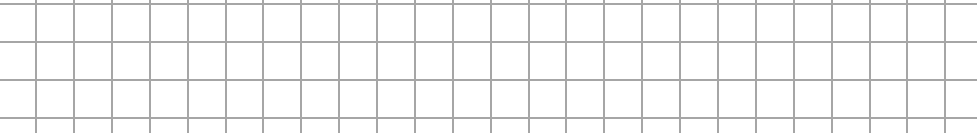
(Suggested maximum time: 15 minutes)

A GAA club runs a weekly club lotto.

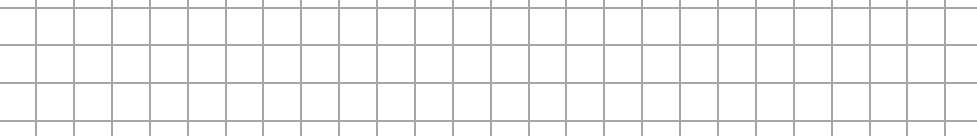
The club sells €400 worth of tickets every week.



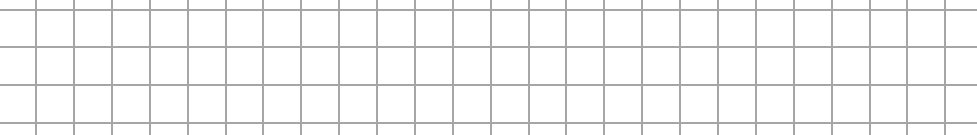
- (a) Each week, 35% of the lotto ticket sales is added to the jackpot. Work out how much the jackpot increases by each week if it is not won.



- (b)** On top of the contribution to the jackpot, there is a weekly cost of €50 to run the lotto. Work out the **weekly profit** made by the club from the lotto ticket sales.



- (c) Work out how many weeks it will take the club to make €1800 profit from the lotto.



(d) Work out the percentage profit made by the club each week from lotto ticket sales.

% Profit = $\frac{\text{Profit}}{\text{Total Ticket Sales}}$

(e) Donal won the jackpot when it reached €5000.
 He decides to invest his winnings at a rate of 2% for 2 years.
 Find the value of Donal’s investment at the end of the 2 years.

$F = P(1 + i)^t$

Label any extra work clearly with the question number and part.

[illegible]

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Time: 2 hours



⑤ 2024-J119-1-EL-20/20